URBAN AIR-QUALITY CONSISTENCY PROFILING

with Relaxed Functional Dependencies

A reproducible analytical workflow for approximate environmental rules

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KDIID · Final project presentation

Consistency—not prediction, not causality

Why deterministic rules are the wrong target

Urban environmental data are:

- noisy and affected by measurement uncertainty
- variable across time, sites, and weather conditions
- influenced by factors that are not fully observed

The analytical question

When observations are similar in context and pollutant levels, how often do other measurements remain similar?

Measure approximate regularity—without forcing exact laws.

The goal is to quantify approximate consistency, not to predict outcomes or prove causality.

A conceptual Urban Digital Twin

The Digital Twin is a framing device for an analytical loop:

01
HISTORICAL DATA
Two stations, pollutants, meteorology

02
APPROXIMATE RULES
RFDs describe expected similarity

03
VIOLATIONS
Cases that break the expected pattern

04
MONITORING
Monthly metrics reveal regime changes

Not built here 3D platform · real-time infrastructure · operational control system

The contribution is a reproducible analytical workflow—not a 3D or real-time platform.

Dataset design: full profiling, balanced RFD sample

Beijing Multi-Site Air Quality Data Set

Aotizhongxin + Changping · March 2013–February 2017

66,619	11	1,500	1,124,250
clean observations	final variables	balanced RFD rows	pairwise comparisons

Missing-value policy summarize missingness → drop incomplete rows on selected columns → no imputation

Use all cleaned observations for profiling; reduce only the quadratic pairwise experiments.

From exact equality to expected similarity

CLASSICAL FD

$$X_1 = X_2$$

implies

$$Y_1 = Y_2$$

exact constraints

RELAXED FD

$$X_1 \approx X_2$$

implies expected similarity

$$Y_1 \approx Y_2$$

threshold-based regularities

RFDs match noisy environmental observations better than deterministic functional dependencies.

How similarity is defined

CATEGORICAL ATTRIBUTES

station, time_slot → exact match

TIME

hour → circular distance ≤ 1 hour

NUMERICAL ATTRIBUTES

absolute distance thresholds

MEDIUM SETUP · MAIN CONFIGURATION

TEMP	≤ 2.0	PM10	≤ 15.0
DEWP	≤ 2.0	NO2	≤ 10.0
WSPM	≤ 1.0	O3	≤ 10.0
PM2.5	≤ 10.0		

Strict and relaxed settings are used as sensitivity analysis.

> **Similarity is manually defined through interpretable thresholds, then checked under strict, medium and relaxed settings.**

Four metrics keep the interpretation honest

SUPPORT

antecedent pairs / all pairs

How often are tuples comparable?

CONFIDENCE

valid pairs / antecedent pairs

How often does the rule hold?

BASELINE

mean confidence over 30 seeded RHS permutations

What happens by chance?

LIFT

$$\text{LIFT} = \frac{\text{confidence}}{\text{baseline confidence}}$$

> 1 means more informative than chance

Lift prevents moderate confidence from being mistaken for weak evidence—or overvalued in isolation.

One pipeline, multiple checks on robustness

01 Preprocess → 02 Profile → 03 Candidate RFDs → 04 Permutation baseline → 05 Bootstrap → 06 Temporal split → 07 Monthly profiling → 08 Violations

DESCRIBE

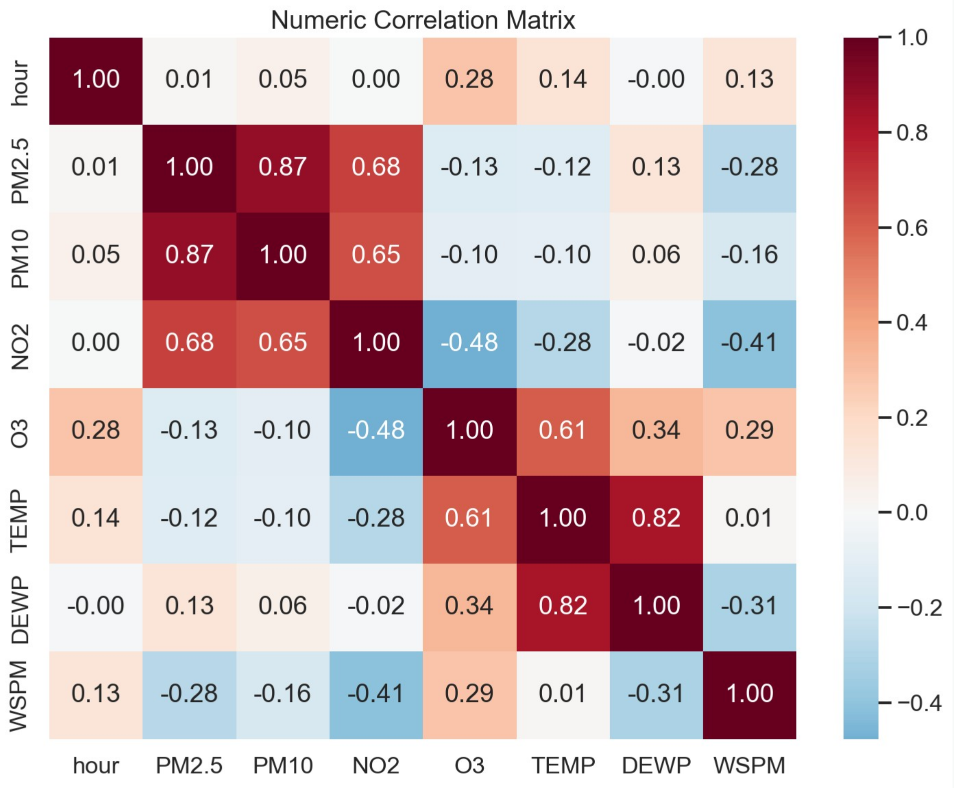
COMPARE WITH CHANCE

TEST STABILITY

INVESTIGATE

> The experiment tests robustness and stability—not only whether a single rule looks plausible.

Descriptive profiling reveals one dominant pattern



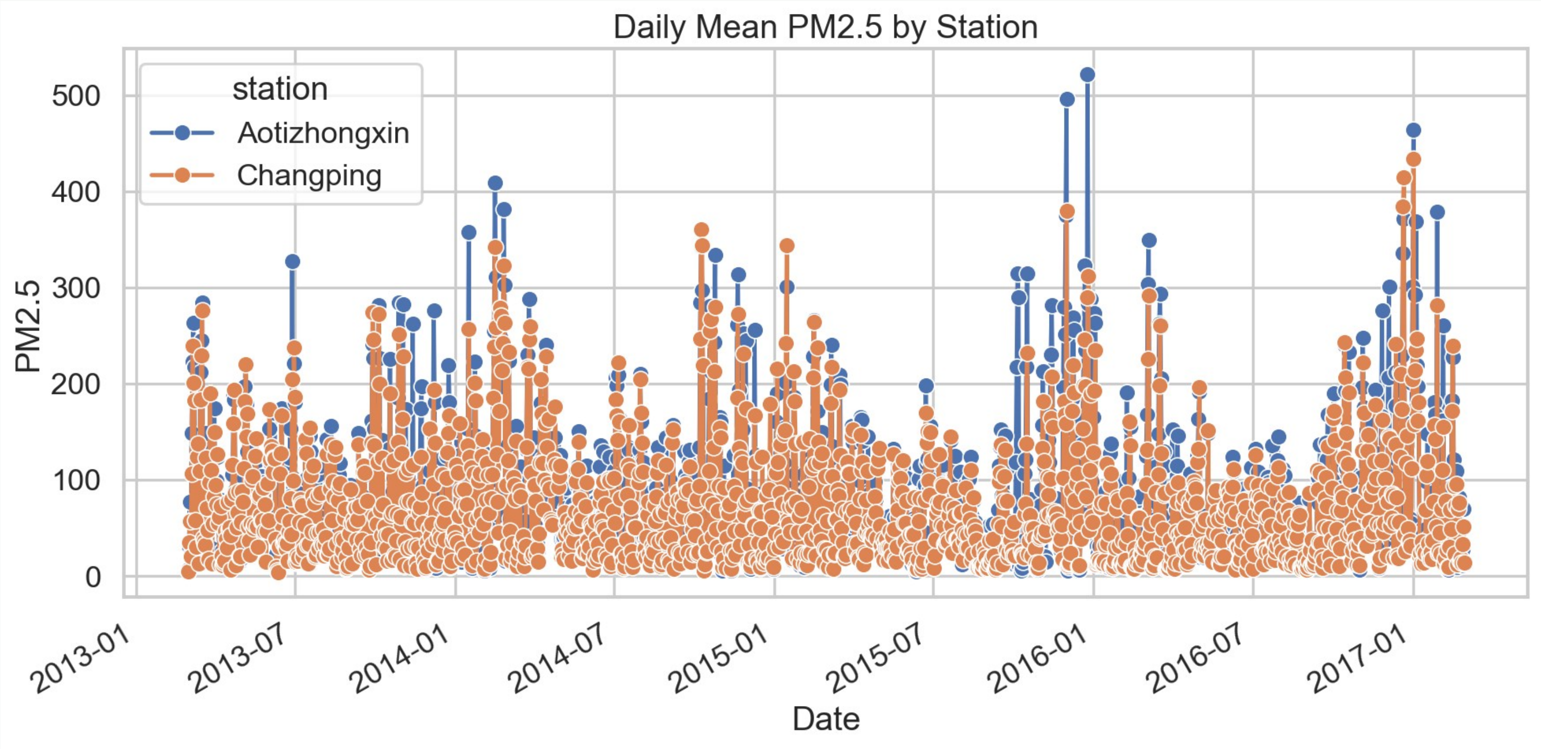
0.874

PM2.5–PM10
Pearson correlation

Correlation is descriptive context—not an RFD metric and not causal evidence.

Fine and coarse particulate matter form the clearest descriptive relationship in the dataset.

PM2.5 behavior changes across time and stations



Temporal and site-level variability motivates continuous profiling rather than one static summary.

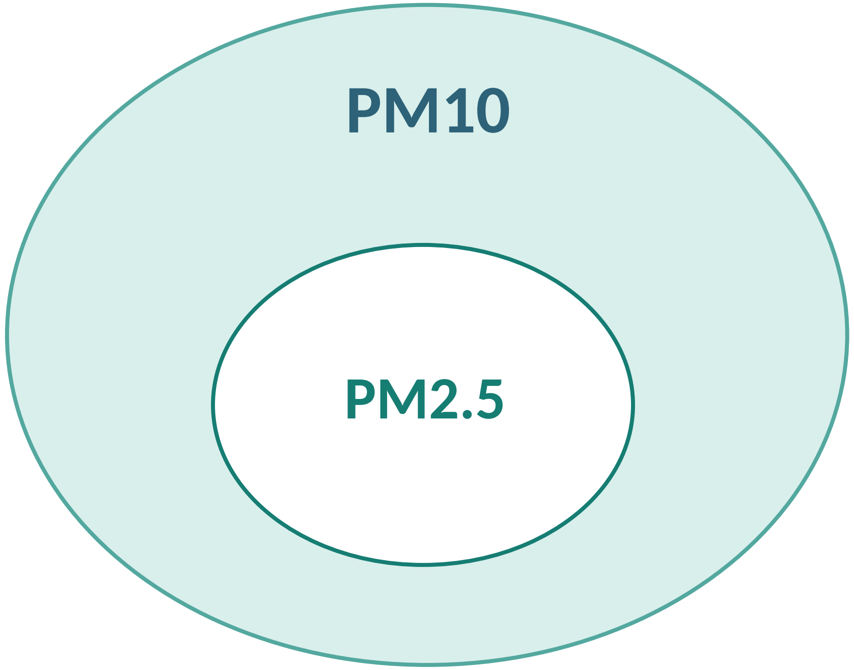
Raw RFDs: moderate confidence, strong lift

Rule	Support	Conf.	Baseline	Lift
station, PM2.5, NO2 → PM10	**0.025**	**0.483**	**0.149**	**3.24**
station, time_slot, PM2.5 → PM10	0.018	0.449	0.148	**3.04**
station, PM2.5 → PM10	0.073	0.434	0.147	**2.94**

3.24× above permutation baseline

Confidence is not deterministic, but the particulate rules are roughly three times above chance.

Environmental meaning: coherence is not causality



Fine particles are included within the broader PM10 size fraction.

- Similar PM2.5 at the same station often accompanies similar PM10.
- NO2 may capture shared emission conditions or common context.
- Meteorology and particle composition can still alter the relationship.

COHERENCE ≠ CAUSALITY

The rule describes consistency between measurements; it does not identify a causal mechanism.

Discretized RFDs trade granularity for confidence

station, PM2.5_bin, NO2_bin → PM10_bin

0.079	0.689	0.333	2.07
support	confidence	baseline	lift

Low / medium / high quantile classes create broader equivalence groups.

Higher confidence is expected; raw and binned rules do not impose the same severity.

> **Binning supports qualitative profiling, but the easier comparison must not be confused with stronger raw evidence.**

The top particulate rule remains relatively stable

BOOTSTRAP · 30 BALANCED RESAMPLES

0.486
mean confidence

3.35
mean lift

95% empirical lift interval
[3.15, 3.57]

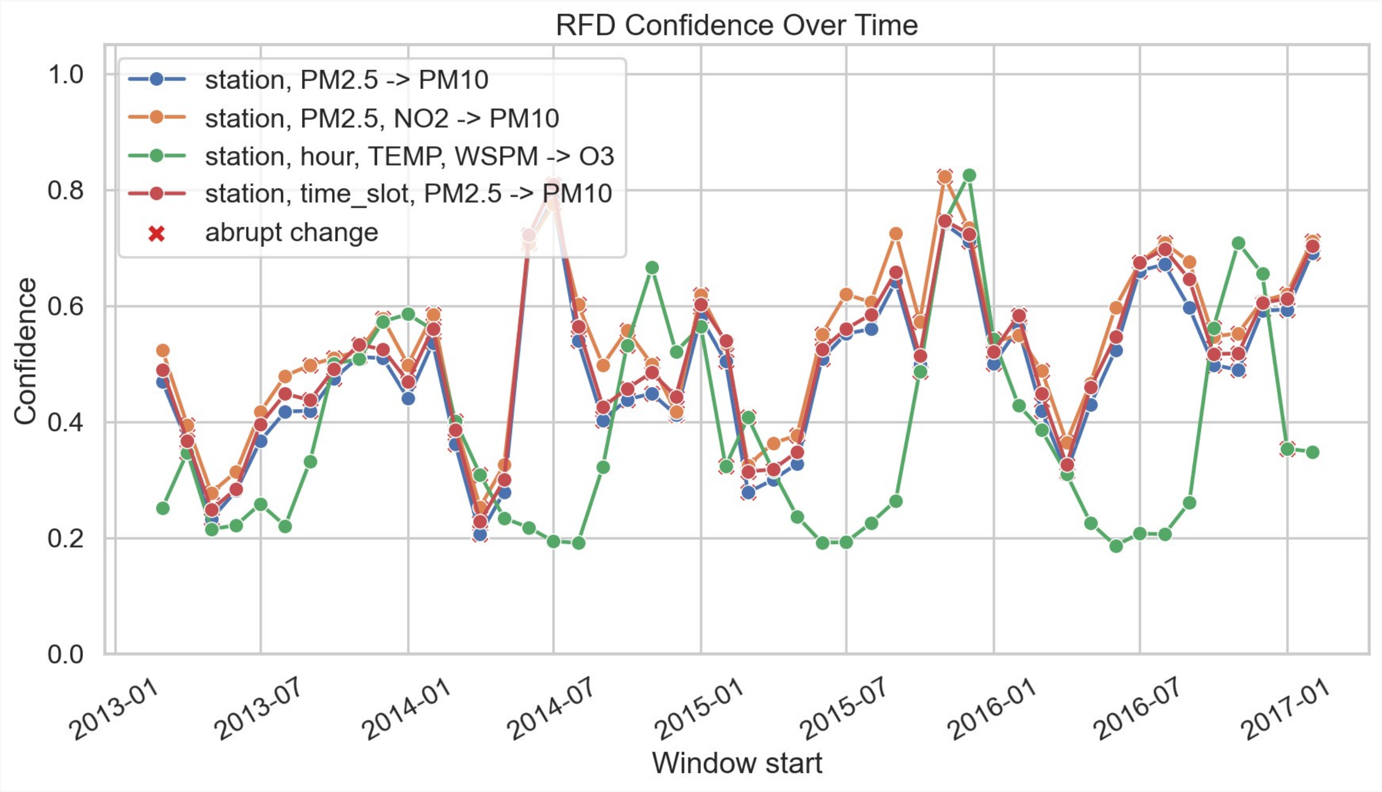
CHRONOLOGICAL VALIDATION · 75 / 25

TRAIN	TEST
Mar 2013–Feb 2016	Mar 2016–Feb 2017
lift 3.15	**lift 3.35**

Confidence also rises from 0.444 to 0.528.

Resampling and the held-out year both preserve a lift above three for the top rule.

Continuous profiling exposes changing regimes



48

monthly windows

- confidence varies over time
- support changes with comparability
- pair counts affect reliability

Monthly RFD metrics can flag regime changes—but confidence must be read with support and pair counts.

Violations are investigation targets—not just errors

A violation may reflect:

- local site heterogeneity
- unobserved emission events
- meteorology outside the rule
- different particulate composition
- measurement or data-quality issues

Exported strongest pairs make the analysis traceable to concrete observations.

The value of violations lies in targeted follow-up, not automatic labeling.

What this project contributes

- A reproducible workflow from preprocessing to violation export.
- **Not causal inference: RFDs profile consistency, not mechanisms.**
- Permutation baselines and lift support a defensible reading of confidence.
- Monthly profiling connects the method to data-stream monitoring.
- **0 rules above support ≥ 0.01 and confidence ≥ 0.85: a conservative result, not a failure.**

> **“The contribution is not to find strong deterministic rules, but to measure how much approximate consistencies are more informative than chance—and how stable they remain over time.”**

Backup · References

01 **Beijing Multi-Site Air Quality Data Set, UCI Machine Learning Repository**

02 Caruccio, Deufemia, Polese, “Relaxed Functional Dependencies — A Survey of Approaches”, IEEE TKDE, 2016

03 Caruccio, Deufemia, Polese, “Mining relaxed functional dependencies from data”, Data Mining and Knowledge Discovery, 2020